

Pharma Search

Project Report submitted for the partial fulfilment of grade for the course

Data Structures and Algorithms -3 (25CS2103E)

in

A.Y. 2026-2027

Submitted by

M. Geethika chowdary (2520030056)

K. Poojitha sai (2520030103)

R. Alekhya(2520030209)

Team Number: 18

Section: 2

Under the guidance of

Vinay Kumar

Assistant Professor, CSE



Hyderabad-500090, Telangana, India.

ABSTRACT

This project presents a Pharma Search System designed to improve the retrieval and management of pharmaceutical information using advanced Data Structures and Algorithms. As the number of medicine records increases, traditional search methods become slower and less efficient. The proposed system allows users to quickly search for medicines and retrieve information such as medicine names, manufacturers, dosage, uses, and available substitutes.

The system applies advanced string-searching and matching techniques to improve search speed and accuracy. Algorithms such as Knuth-Morris-Pratt (KMP), Rabin-Karp, and Edit Distance can be used for pattern searching and fuzzy matching, allowing the system to identify partial or similar medicine names. Suitable data structures and algorithms are selected according to the search requirements and dataset size.

The project demonstrates the practical application of advanced algorithms in solving real-world pharmaceutical information retrieval problems. By reducing search time and improving accuracy, the Pharma Search System provides an efficient and user-friendly solution for managing and searching medicine information.

Signature of the Guide